1. Global and Local spatial correlation (Global and Local moran I)

Without filtering based on correct evidence

model	N Query	Reply_amb	non_ambig	correct_pr	pct_with_e	TP	FP	accuracy	precision	recall	F1
deepseek/deepseek-chat-v3-0324	74	22	52	30	0.459	15	6	0.405	0.714	0.405	0.517
deepseek/deepseek-r1-0528	74	6	68	52	0.797	27	9	0.703	0.75	0.73	0.74
meta-llama/llama-3.1-8b-instruct	74	22	52	17	0.081	0	1	0.23	0	0	0
openai/gpt-4	74	35	39	12	0.027	0	0	0.162	0	0	0
openai/gpt-4o-mini	74	21	53	29	0.176	12	4	0.392	0.75	0.324	0.453
openai/o3	74	11	63	30	0.365	6	4	0.405	0.6	0.162	0.255
openai/o4-mini	74	1	73	59	0.865	29	7	0.797	0.806	0.784	0.795
qwen/qwen3-32b	74	1	73	39	0.716	16	13	0.527	0.552	0.432	0.485

After filtering based on correct evidence

model	N_Query	Reply_amb	non_ambig	correct_pr	pct_with_e	TP_Eviden	FP_Eviden	accuracy_E	precision_	recall_Evid	F1_Evidence
deepseek/deepseek-chat-v3-0324	74	22	52	18	0.459	10	6	0.243	0.625	0.27	0.377
deepseek/deepseek-r1-0528	74	6	68	49	0.797	27	9	0.662	0.75	0.73	0.74
meta-llama/llama-3.1-8b-instruct	74	22	52	2	0.081	0	1	0.027	0	0	0
openai/gpt-4	74	35	39	2	0.027	0	0	0.027	0	0	0
openai/gpt-4o-mini	74	21	53	9	0.176	8	4	0.122	0.667	0.216	0.327
openai/o3	74	11	63	17	0.365	6	4	0.23	0.6	0.162	0.255
openai/o4-mini	74	1	73	56	0.865	29	7	0.757	0.806	0.784	0.795
qwen/qwen3-32b	74	1	73	35	0.716	16	13	0.473	0.552	0.432	0.485

2. Bivariate spatial correlation

Without filtering based on correct evidence

model	N Query	Reply_amb	non_ambig	correct_pr	pct_with_e	TP	FP	accuracy	precision	recall	F1
deepseek/deepseek-chat-v3-0324	50	10	40	31	0.86	31	9	0.62	0.466	0.407	0.43
deepseek/deepseek-r1-0528	50	7	43	29	0.58	29	14	0.58	0.419	0.406	0.412
meta-llama/llama-3.1-8b-instruct	50	15	35	23	0.28	23	12	0.46	0.256	0.197	0.222
openai/gpt-4	50	22	28	18	0.52	18	10	0.36	0.413	0.182	0.243
openai/gpt-4o-mini	50	5	45	31	0.54	31	14	0.62	0.43	0.293	0.32
openai/o3	50	0	50	35	0.76	35	15	0.7	0.597	0.328	0.343
openai/o4-mini	50	0	50	32	1	32	18	0.64	0.489	0.387	0.43
qwen/qwen3-32b	50	1	49	14	0.98	14	35	0.28	0.244	0.436	0.182

After filtering based on correct evidence

model	N_Query	Reply_amb	non_ambig	correct_pr	pct_with_e	TP_Eviden	FP_Eviden	accuracy_E	precision_	recall_Evid	F1_Evidence
deepseek/deepseek-chat-v3-0324	50	10	40	30	0.86	30	8	0.6	0.475	0.399	0.428
deepseek/deepseek-r1-0528	50	7	43	19	0.58	19	10	0.38	0.428	0.321	0.36
meta-llama/llama-3.1-8b-instruct	50	15	35	8	0.28	8	6	0.16	0.296	0.068	0.111
openai/gpt-4	50	22	28	17	0.52	17	9	0.34	0.421	0.174	0.238
openai/gpt-4o-mini	50	5	45	19	0.54	19	8	0.38	0.467	0.191	0.264
openai/o3	50	0	50	25	0.76	25	13	0.5	0.591	0.242	0.295
openai/o4-mini	50	0	50	32	1	32	18	0.64	0.489	0.387	0.43
qwen/qwen3-32b	50	1	49	13	0.98	13	35	0.26	0.238	0.427	0.173

3. GetISord (Hotspot detection)

Without filtering based on correct evidence

model	N Query	Reply_amb	n_implaus	non_ambig	correct_pr	pct_with_e	TP-Borough level	FP_ Borough level	accuracy	precision	recall	F1
deepseek/deepseek-chat-v3-0324	21	7	1	13	3	0.381	23	10	0.921	0.697	0.338	0.455
deepseek/deepseek-r1-0528	21	0	0	21	20	0.857	67	0	0.999	1	0.985	0.992
meta-llama/llama-3.1-8b-instruct	21	10	0	11	2	0.048	0	5	0.895	0	0	0
openai/gpt-4	21	10	2	9	2	0.19	2	3	0.9	0.4	0.029	0.054
openai/gpt-4o-mini	21	0	5	16	1	0.429	15	14	0.903	0.517	0.221	0.31
openai/o3	21	0	0	21	5	0.143	12	3	0.915	0.8	0.176	0.289
openai/o4-mini	21	0	0	21	21	0.857	68	0	1	1	1	1
qwen/qwen3-32b	21	0	0	21	5	0.429	19	7	0.919	0.731	0.279	0.404

After filtering based on correct evidence

model	N_Query	Reply_amb	n_implaus	non_ambig	correct_predicted_Evidence	pct_with_e	TP_Evidence_Borough level	FP_Evidence_Borough level	accuracy_l	precision_l	recall_Evidenc	F1_Evidence
deepseek/deepseek-chat-v3-0324	21	7	1	13	3	0.381	23	10	0.921	0.697	0.338	0.455
deepseek/deepseek-r1-0528	21	0	0	21	20	0.857	67	0	0.999	1	0.985	0.992
meta-llama/llama-3.1-8b-instruct	21	10	0	11	2	0.048	0	5	0.895	0	0	0
openai/gpt-4	21	10	2	9	2	0.19	2	3	0.9	0.4	0.029	0.054
openai/gpt-4o-mini	21	0	5	16	1	0.429	15	14	0.903	0.517	0.221	0.31
openai/o3	21	0	0	21	5	0.143	12	3	0.915	0.8	0.176	0.289
openai/o4-mini	21	0	0	21	21	0.857	68	0	1	1	1	1
qwen/qwen3-32b	21	0	0	21	5	0.429	19	7	0.919	0.731	0.279	0.404

4. Spatial outlier analysis

Without filtering based on correct evidence

model	N Query	Reply_amb	n_implaus	non_ambig	correct_pr	pct_with_e	TP	FP	accuracy	precision	recall	F1
deepseek/deepseek-chat-v3-0324	50	2	0	48	46	1	46	2	0.92	0.833	0.938	0.856
deepseek/deepseek-r1-0528	50	0	1	49	46	1	46	3	0.92	0.821	0.943	0.853
meta-llama/llama-3.1-8b-instruct	50	29	0	21	7	0.095	7	14	0.14	0.271	0.107	0.134
openai/gpt-4	50	37	0	13	13	1	13	0	0.26	1	0.319	0.458
openai/gpt-4o-mini	50	10	0	40	33	0.75	33	7	0.66	0.907	0.633	0.711
openai/o3	50	5	0	45	31	0.933	31	14	0.62	0.758	0.774	0.68
openai/o4-mini	50	0	0	50	49	0.98	49	1	0.98	0.984	0.988	0.986
qwen/qwen3-32b	50	3	0	47	27	0.979	27	20	0.54	0.515	0.645	0.476

After filtering based on correct evidence

model	N_Query	Reply_amb	n_implaus	non_ambig	correct_predicted_Evidence	pct_with_e	TP_Evidence	FP_Evidence	accuracy_l	precision_l	recall_Evidenc	F1_Evidence
deepseek/deepseek-chat-v3-0324	50	2	0	48	46	1	46	2	0.92	0.833	0.938	0.856
deepseek/deepseek-r1-0528	50	0	1	49	46	1	46	3	0.92	0.821	0.943	0.853
meta-llama/llama-3.1-8b-instruct	50	29	0	21	0	0.095	0	2	0	0	0	0
openai/gpt-4	50	37	0	13	13	1	13	0	0.26	1	0.319	0.458
openai/gpt-4o-mini	50	10	0	40	29	0.75	29	1	0.58	0.978	0.567	0.711
openai/o3	50	5	0	45	29	0.933	29	13	0.58	0.756	0.74	0.669
openai/o4-mini	50	0	0	50	49	0.98	49	0	0.98	1	0.988	0.994
qwen/qwen3-32b	50	3	0	47	27	0.979	27	19	0.54	0.565	0.645	0.479

5. Spatial Distance decay analysis

Without filtering based on correct evidence

model	N Query	Reply_amb	n_implaus	non_ambig	correct_pr	pct_with_e	TP	FP	accuracy	precision	recall	F1
deepseek/deepseek-chat-v3-0324	50	7	0	43	43	1	43	0	0.86	0.667	0.561	0.608
deepseek/deepseek-r1-0528	50	0	0	50	50	1	50	0	1	0.667	0.667	0.667
meta-llama/llama-3.1-8b-instruct	50	6	6	38	36	0.947	36	2	0.72	0.627	0.483	0.545
openai/gpt-4	50	16	0	34	34	1	34	0	0.68	0.667	0.456	0.541
openai/gpt-4o-mini	50	1	4	45	45	1	45	0	0.9	0.667	0.606	0.634
openai/o3	50	2	26	22	18	0.818	18	4	0.36	0.539	0.239	0.33
openai/o4-mini	50	0	2	48	48	1	48	0	0.96	0.667	0.644	0.655
qwen/qwen3-32b	50	0	12	38	38	1	38	0	0.76	0.667	0.511	0.578

After filtering based on correct evidence

model	N_Query	Reply_amb	n_implaus	non_ambig	correct_predicted_Evidence	pct_with_e	TP_Evidence	FP_Evidence	accuracy_l	precision_l	recall_Evidenc	F1_Evidence
deepseek/deepseek-chat-v3-0324	50	7	0	43	43	1	43	0	0.86	0.667	0.561	0.608
deepseek/deepseek-r1-0528	50	0	0	50	50	1	50	0	1	0.667	0.667	0.667
meta-llama/llama-3.1-8b-instruct	50	6	6	38	36	0.947	36	0	0.72	0.667	0.483	0.56
openai/gpt-4	50	16	0	34	34	1	34	0	0.68	0.667	0.456	0.541
openai/gpt-4o-mini	50	1	4	45	45	1	45	0	0.9	0.667	0.606	0.634
openai/o3	50	2	26	22	17	0.818	17	1	0.34	0.639	0.222	0.328
openai/o4-mini	50	0	2	48	48	1	48	0	0.96	0.667	0.644	0.655
qwen/qwen3-32b	50	0	12	38	38	1	38	0	0.76	0.667	0.511	0.578